

Question				Marking details	Marks Available																		
					AO1	AO2	AO3	Total	Maths	Prac													
9	(a)	(i)		Straight line through origin shows a proportional relationship (1) so true for thinking distance but not braking distance (1) OR Braking distance is a curve so it isn't true (1) Thinking distance is a straight line through the origin so true (1) Accept answers based on data			2	2															
		(ii)		40 mph converted to 18 m/s (1) Distance of 12 [m] seen anywhere (1) Answer = 0.67 [s] (1) Award 2 marks for an answer of 0.3 [s]	1	1 1		3	3														
		(iii)		<table><tr><td>Speed (mph)</td><td>0</td><td>20</td><td>30</td><td>40</td><td>60</td><td>70</td></tr><tr><td>Total stopping distance (m)</td><td>0</td><td>12</td><td>22 or 22.5 or 23</td><td>36</td><td>72 or 72.5 or 73</td><td>95 or 95.5 or 96</td></tr></table> 5 or 6 correct (2) 3 or 4 correct (1) 1 or 2 correct (0)	Speed (mph)	0	20	30	40	60	70	Total stopping distance (m)	0	12	22 or 22.5 or 23	36	72 or 72.5 or 73	95 or 95.5 or 96		2	2	2	
Speed (mph)	0	20	30	40	60	70																	
Total stopping distance (m)	0	12	22 or 22.5 or 23	36	72 or 72.5 or 73	95 or 95.5 or 96																	
		(iv)		5 points plotted correctly (ignore (0,0) $\pm$ 1 small square (2) ecf on table 4 points plotted correctly $\pm$ 1 small square (1) 3 or less points plotted correctly $\pm$ 1 small square (0) Best fit curve through (0,0) $\pm$ 1 small square (1) Don't accept double, thick, disjointed, wispy curves		3		3	3														
	(b)			At 30 mph stopping distance = 22 or 22.5 m or 23 m (ecf) (1) At 20 mph the stopping distance = 12 m (ecf) (1) 12 m is less than 15 m or 12 m is less than the distance after the bend or there is a 3 m gap so less chance of a collision (1)			3	3															
				Question 9 total	1	7	5	13	8	0													